TYPES OF REACTIONS

Name: ..... **Date:** .....

Part 1: use the patterns in the box to classify each equation. S = synthesis, D = decomposition, SR = single replacement, DR = double replacement, C = combustion. Part 2: answer in full sentences.

THE FIVE TYPES

Synthesis (S): two or more substances join to make one. A + B → AB

Decomposition (D): one substance breaks down into two or more. AB → A + B

Single replacement (SR): one element swaps places with another in a compound. A + BC → AC + B

Double replacement (DR): two compounds swap partners. AB + CD → AD + CB

Combustion (C): a fuel burns in oxygen; a hydrocarbon gives carbon dioxide and water. fuel + O_2 → CO_2 + H_2O

PART 1: CLASSIFY EACH REACTION

1. $2H_2 + O_2 \rightarrow 2H_2O$ S D SR DR C
2. $2H_2O \rightarrow 2H_2 + O_2$ S D SR DR C
3. $CH_4 + 2O_2 \rightarrow CO_2 + 2H_2O$ S D SR DR C
4. $Zn + 2HCl \rightarrow ZnCl_2 + H_2$ S D SR DR C
5. $NaCl + AgNO_3 \rightarrow NaNO_3 + AgCl$ S D SR DR C
6. $CaCO_3 \rightarrow CaO + CO_2$ S D SR DR C
7. $2Mg + O_2 \rightarrow 2MgO$ S D SR DR C
8. $Fe + CuSO_4 \rightarrow FeSO_4 + Cu$ S D SR DR C
9. $C_3H_8 + 5O_2 \rightarrow 3CO_2 + 4H_2O$ S D SR DR C
10. $2KI + Pb(NO_3)_2 \rightarrow PbI_2 + 2KNO_3$ S D SR DR C
11. $2Na + Cl_2 \rightarrow 2NaCl$ S D SR DR C
12. $2KClO_3 \rightarrow 2KCl + 3O_2$ S D SR DR C

PART 2: THINK ABOUT IT

Write your own example of a decomposition reaction as a word equation.

Which two reaction types are the reverse of each other? Explain, using equations 1 and 2.

TYPES OF REACTIONS ANSWER KEY

Name: ..... **Date:** .....

PART 1: CLASSIFY EACH REACTION

1. S 2. D 3. C 4. SR 5. DR 6. D 7. S 8. SR 9. C 10. DR 11. S 12. D

PART 2: THINK ABOUT IT

1. Sample: copper carbonate → copper oxide + carbon dioxide (any AB → A + B example). 2. Synthesis and decomposition. In equation 1 hydrogen and oxygen join to make water; in equation 2 water breaks apart into hydrogen and oxygen, the same equation read backwards.

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